Report on the Use of NotebookLM AI Tool in an Academic Task

1. Task Description

Selected Task

To summarize and analyze three academic sources in English on the topic of “Evaluating AI Tools in Engineering Education,” and to prepare an executive summary in Arabic that can serve as a quick reference for future academic research.

Reason for Selection

This task represents a realistic academic challenge involving the processing of a large amount of information from multiple sources, extracting key ideas, and organizing them for easy future reference. The task requires a deep understanding of the content and the ability to connect different sources, areas where AI tools can provide valuable support, with human judgment remaining essential for editing and verification.

2. AI Tool

Tool Used: Google NotebookLM

Reason for Selection

NotebookLM was chosen because it relies solely on the sources uploaded by the user, which significantly reduces hallucinations and provides documented answers with accurate citations. The tool also offers advanced capabilities for summarizing long documents and extracting key points, with the ability to trace information back to its original source. Furthermore, it can compare information from multiple sources and identify points of agreement and disagreement.

3. AI Workflow

Original Workflow

The task required reading three complete research sources (each ranging from 10-20 pages) and manually extracting information. These included: a study from Nature Scientific Reports on

AI tools in mechanical engineering education, a study from ASEE on developing critical thinking, and a paper from IFLA on a comparative analysis of AI tools to support academic research.

Prompts Used

1. **First Prompt (Source Upload):** Analyze the research and extract the objectives, methodology, main results, and recommendations from each study.
2. **Second Prompt (Comparison and Synthesis):** Identify common themes and differences between the studies, and synthesize the main recommendations.
3. **Third Prompt (Preparation for Arabic):** Prepare an executive summary in Arabic including an introduction, key findings, and practical applications, with clear citations.

AI Output (Initial Summary)

The tool identified four main common themes: (1) varying performance among tools, with Copilot achieving the highest accuracy (60.38%) followed by Gemini (57.13%) and then ChatGPT (46.63%); (2) tools excelling in theoretical questions but struggling with numerical problems; (3) the potential to enhance critical thinking through structured educational integration; (4) Gemini's superiority in supporting academic research compared to Copilot. It also identified points of difference, such as Copilot being ranked higher in accuracy in the Nature study versus Gemini's preference in supporting research according to the IFLA study.

Final Version After Editing and Review

Executive Summary - Evaluation of AI Tools in Academic Education and Research

The past few years have witnessed a widespread proliferation of generative AI tools such as ChatGPT, Google Gemini, and Microsoft Copilot in academic and educational fields. This summary presents the main findings extracted from three recent research studies.

The studies showed significant variation in the accuracy of answers among the tools, with Microsoft Copilot achieving the highest accuracy rate (60.38%) in solving engineering questions, followed by Google Gemini (57.13%), and then ChatGPT (46.63%). A comparative study also showed that Gemini provides stronger support for academic research and information literacy education compared to Copilot. The studies agree that AI tools perform exceptionally well in theoretical questions and editorial tasks but struggle with numerical problems and complex equations. A study from ASEE also showed that the structured integration of these tools into educational assignments can contribute to developing students' critical thinking skills when asked to compare and critically evaluate outputs. The studies point to key challenges, most notably over-reliance which may weaken problem-solving skills, concerns about academic dishonesty, and limited reliability in computational tasks.

Recommendations: For researchers, it is recommended to use NotebookLM as a specialized tool for processing academic documents. For educators, design assignments that require the use of AI as part of the learning process, focusing on prompt engineering skills and output verification. For students, exercise caution when using AI in numerical problems and rely on personal understanding.

My Edits: I restructured the text, improved the linguistic accuracy of the Arabic translation, added points about challenges, organized recommendations into clear categories, and verified citations.

4. Critical Thinking

How the Tool Improved Productivity

The tool reduced the time required to read and analyze the three sources from approximately 4-6 hours to less than an hour by automatically extracting key points. It helped me organize information from multiple sources in a coordinated manner and handled the routine task of extracting essential information, allowing me to focus on deep analysis. It also provided an initial translation of the content into Arabic, saving significant time in drafting the executive summary.

Strengths and Limitations

Strengths: Accuracy and reliability due to its reliance solely on uploaded sources, embedded citations that allow tracing every piece of information to its source, and excellent ability to summarize long texts without losing essential information.

Limitations: The quality of outputs depends entirely on the quality of the uploaded sources, it cannot provide deep critical analysis, the Arabic translation required linguistic review, and the tool struggled to synthesize information from disparate sources without human intervention.

Changes Made After Review

I rephrased the language to be more formal and appropriate for an academic report, added a comparative analysis between the results of different studies, organized the recommendations more clearly, and verified the information, figures, and citations by comparing them with the original texts.

Importance of Human Judgment

This experience highlights that human judgment is indispensable: humans are capable of evaluating the quality of AI outputs and determining their suitability for the context. While the tool excels at extracting superficial information, deep analysis and connecting ideas still require human expertise. Machine translation needs human review to ensure accuracy, and the researcher is responsible for the accuracy of the final content. The tool cannot understand specific needs or the particular context of the task without precise human guidance.

Acknowledgement of AI Tool Use I acknowledge that this report was prepared with the assistance of Google NotebookLM for summarizing academic sources and extracting essential information, with all editing, review, and critical analysis performed entirely by a human. All citations have been verified for accuracy by referring to the original sources.